

## Table of Contents

| Chapter No. | Chapter Title                    | Questions Included (Range) |
|-------------|----------------------------------|----------------------------|
| 1           | Number Basics                    | 1–15                       |
| 2           | Number Theory                    | 16–30                      |
| 3           | Pattern Printing                 | 31–45                      |
| 4           | Arrays                           | 46–60                      |
| 5           | Strings                          | 61–75                      |
| 6           | Recursion                        | 76–85                      |
| 7           | Sorting & Searching Algorithms   | 86–92                      |
| 8           | Miscellaneous & Logical Problems | 93–100                     |

## **Chapter 1: Number Basics (1–15)**

1. Check if a number is positive, negative, or zero
  2. Find the maximum of three numbers
  3. Check if a number is even or odd
  4. Check whether a year is a leap year
  5. Check if a number is prime
  6. Find the factorial of a number
  7. Generate multiplication table of a number
  8. Print all factors of a number
  9. Count digits in a number
  10. Reverse a number
  11. Sum of digits of a number
  12. Check if a number is palindrome
  13. Check if a number is Armstrong
  14. Calculate power of a number
  15. Calculate simple interest
- 

## **Chapter 2: Number Theory (16–30)**

16. Find GCD of two numbers
17. Find LCM of two numbers
18. Check for a strong number
19. Check if a number is perfect
20. Check for a Harshad number
21. Check if a number is neon
22. Convert decimal to binary
23. Convert binary to decimal
24. Convert decimal to octal
25. Convert decimal to hexadecimal
26. Sum of first N natural numbers

- 27. Sum of squares of first N natural numbers
  - 28. Check if a number is automorphic
  - 29. Find nth term of an arithmetic progression
  - 30. Check for happy number
- 

### **Chapter 3: Pattern Printing (31–45)**

- 31. Print a square star pattern
  - 32. Print right-angled triangle pattern
  - 33. Print pyramid pattern
  - 34. Print inverted pyramid
  - 35. Print Pascal's triangle
  - 36. Print Floyd's triangle
  - 37. Print number triangle
  - 38. Print diamond pattern
  - 39. Print alphabet pattern A to Z
  - 40. Print hollow square pattern
  - 41. Print cross (X) pattern
  - 42. Print zigzag pattern
  - 43. Print butterfly pattern
  - 44. Print heart pattern
  - 45. Print rhombus pattern
- 

### **Chapter 4: Arrays (46–60)**

- 46. Find largest element in an array
- 47. Find smallest element in an array
- 48. Sum of array elements
- 49. Copy one array to another
- 50. Reverse an array
- 51. Count even and odd elements in array

- 52. Find frequency of elements
  - 53. Sort array in ascending order
  - 54. Sort array in descending order
  - 55. Merge two arrays
  - 56. Find duplicate elements in array
  - 57. Find unique elements in array
  - 58. Left rotate an array
  - 59. Right rotate an array
  - 60. Second largest element in array
- 

### **Chapter 5: Strings (61–75)**

- 61. Count vowels and consonants
  - 62. Check if a string is palindrome
  - 63. Find length of a string
  - 64. Convert lowercase to uppercase
  - 65. Reverse a string
  - 66. Check anagram strings
  - 67. Remove vowels from string
  - 68. Count words in a sentence
  - 69. Find duplicate characters
  - 70. Remove duplicate characters
  - 71. Find frequency of characters
  - 72. Sort characters of a string
  - 73. Replace character in string
  - 74. Concatenate two strings
  - 75. Toggle characters of string
- 

### **Chapter 6: Recursion (76–85)**

- 76. Factorial using recursion

- 77. Fibonacci using recursion
  - 78. GCD using recursion
  - 79. Power using recursion
  - 80. Sum of digits using recursion
  - 81. Print array using recursion
  - 82. Reverse string using recursion
  - 83. Palindrome check using recursion
  - 84. Binary search using recursion
  - 85. Tower of Hanoi problem
- 

### **Chapter 7: Sorting & Searching Algorithms (86–92)**

- 86. Linear search
  - 87. Binary search
  - 88. Bubble sort
  - 89. Selection sort
  - 90. Insertion sort
  - 91. Quick sort
  - 92. Merge sort
- 

### **Chapter 8: Miscellaneous & Logical Problems (93–100)**

- 93. Swap two numbers without third variable
  - 94. Generate Fibonacci series
  - 95. Find missing number in array
  - 96. Find first non-repeating character in string
  - 97. Check if two numbers are co-prime
  - 98. Print prime numbers in a range
  - 99. Generate random number
  - 100. Convert Roman numerals to integer
-

## **Chapter 9: Intermediate Logic & Math (101–120)**

101. Convert integer to Roman
102. Find sum of series  $1 + 1/2 + 1/3 + \dots + 1/n$
103. Solve quadratic equation
104. Check magic number
105. Convert number to words
106. Count number of set bits
107. Convert binary to gray code
108. Convert gray code to binary
109. Find power of 2
110. Print prime factors of a number
111. Count trailing zeros in factorial
112. Print nth Fibonacci number
113. Count number of zeroes in a number
114. Find highest power of 2 less than N
115. Convert time (HH:MM) to minutes
116. Calculate digital root
117. Find next greater element in array
118. Find leader elements in array
119. Find missing letter in a sequence
120. Print Pascal triangle using array

---

## **Chapter 10: Advanced Programming Challenges (121–150)**

121. Find subarray with maximum sum (Kadane's Algorithm)
122. Maximum product subarray
123. Longest increasing subsequence
124. Longest common subsequence
125. Longest palindrome substring
126. Check balanced parentheses

127. Convert infix to postfix
128. Evaluate postfix expression
129. Find first repeating element
130. Rotate matrix 90 degrees
131. Spiral print of matrix
132. Transpose of matrix
133. Check symmetric matrix
134. Matrix multiplication
135. Flood fill algorithm
136. Binary search in rotated array
137. Count paths in grid
138. Snake and ladder minimum moves
139. Sudoku solver
140. N-Queens problem
141. Word break problem
142. Maximum depth of binary tree
143. Check BST validity
144. Lowest common ancestor in BST
145. Merge two sorted linked lists
146. Detect cycle in linked list
147. Find intersection point in linked lists
148. Implement stack using queue
149. Implement queue using stack
150. LRU cache implementation